OPEN PROJECT 2026 • SOCIETY OF BUSINESS

# Agentic AI for Dynamic EV Charging Tariff Optimization

A three-agent engine that forecasts demand, recommends dynamic ₹/kWh tariffs, and tracks the outcomes in simulation.

ACN-Data · 30k+ sessions (Caltech/JPL)

UrbanEV · 24,798 piles · 5-min intervals (Shenzhen)

*Demand Prediction Agent → Tariff Pricing Agent → Monitoring & Learning Agent*

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EXECUTIVE SUMMARY

# What we built, and what it delivers

**The problem.** Flat ₹15/kWh pricing ignores real demand — creating peak congestion and idle off-peak chargers.

**Our approach.** A three-agent loop trained on 2.1M+ interval records forecasts utilization one hour ahead, maps forecasts to surge/discount tariffs, then tracks revenue, queues and pricing efficiency to close the loop.

**Headline result.** Under base-case elasticity, the rule-based policy is associated with ~10% higher simulated revenue vs the fixed baseline and a ~2.8-unit lower peak queue proxy per episode.

**+10.25%**

Revenue gain vs ₹15/kWh baseline (base case)

**0.942**

Best-model $R^2$ (Gradient Boosting)

**0.079**

RMSE on 1-hour-ahead utilization

**19.36**

Pricing efficiency (₹/kWh delivered)

*Scope: 2.13M modelled interval-station records · time-ordered 80/20 split. Revenue, queue and pricing-efficiency figures are simulated / proxy outcomes, not live A/B results.*

# Two datasets, one analytical base

## ACN-Data (Caltech / JPL)

- 30k+ real charging sessions; JSON → CSV
- Session granularity: connect/disconnect, kWh, duration
- Used for micro-behaviour: energy & dwell-time profiles

## UrbanEV / ST-EVCDP (Shenzhen)

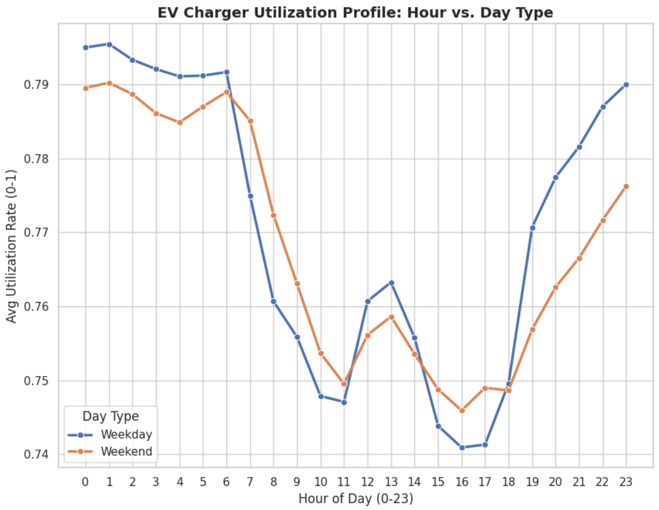
- 24,798 piles · 5-minute interval matrices
- Occupancy, duration, price, station capacity
- Used for temporal & spatial demand + modelling base

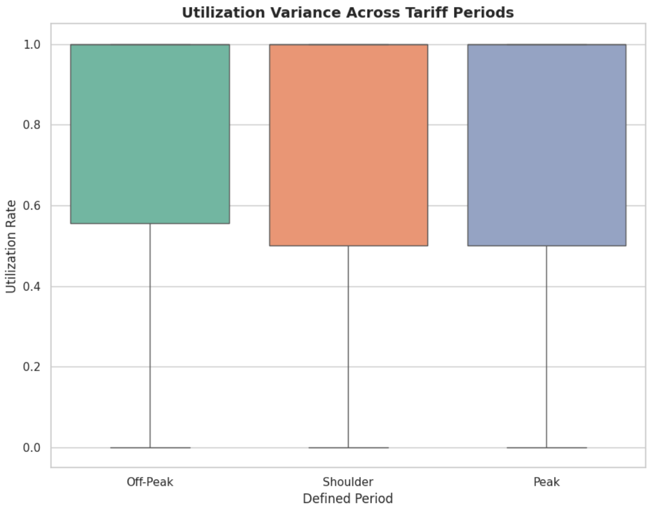
### Pipeline

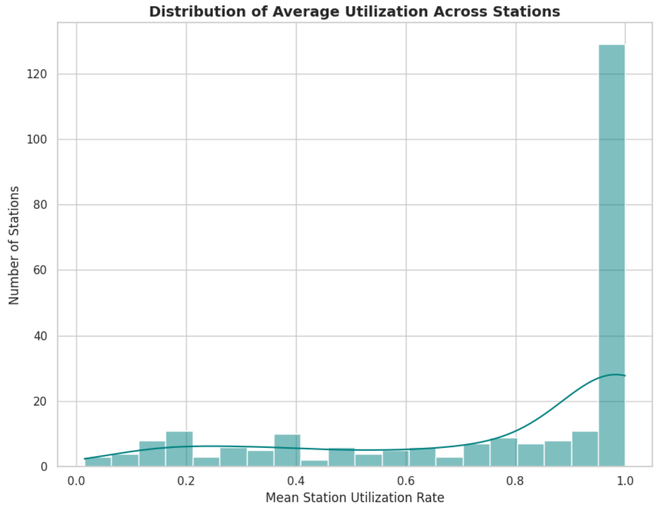
1. Drop missing keys & duplicates →
2. Parse timestamps; build unified index →
3. Clip negatives; forward-fill gaps →
4. Melt matrices → long station-time →
5. Engineer features & utilization rate

*Engineered features: Charger Utilization Rate, Occupancy Density, Queue-Length Proxy, Energy Delivered, Revenue per Session. All assumptions written to a transparency log.*

# Demand behaviour: temporal & session shape




## Data limitation

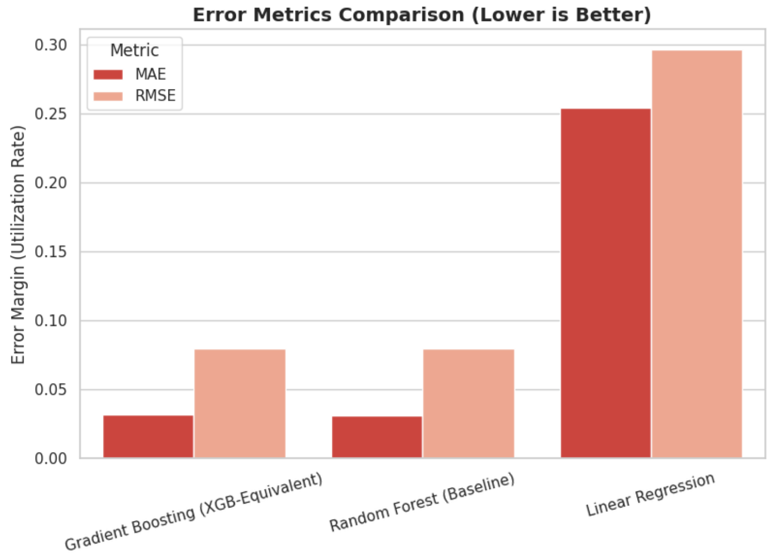
Spatial / geo map omitted — station coordinate merge did not resolve (stations.csv keys). Temporal & station-level views above are unaffected.

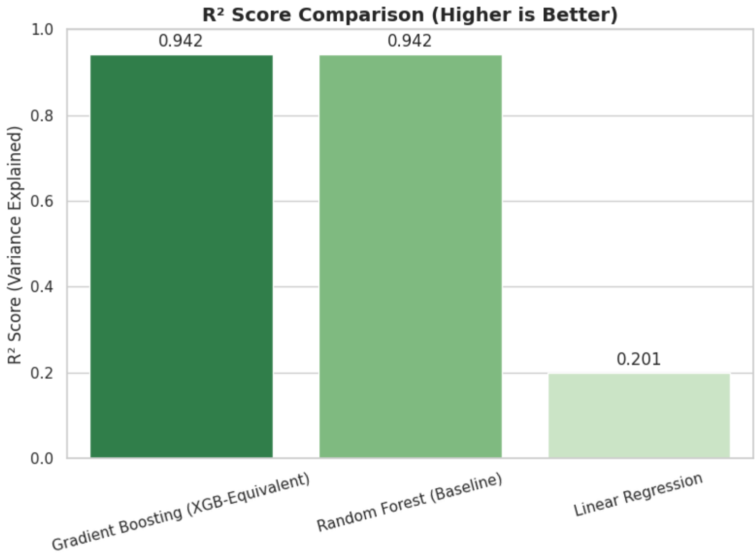
## Key findings

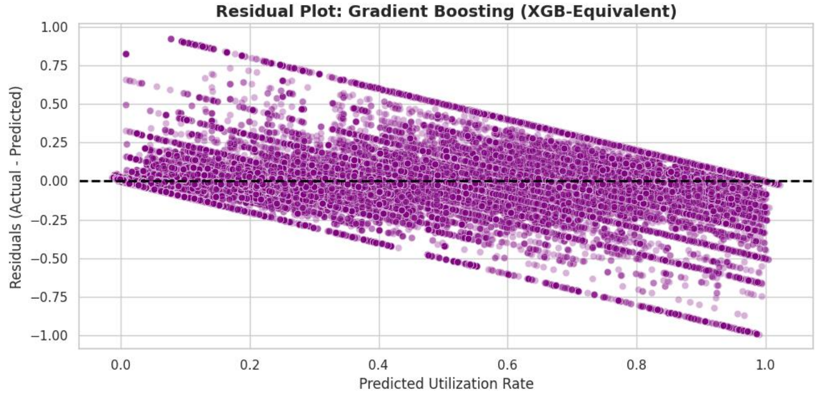
- Strong intraday cycle — utilization dips through mid-day and rises into evening; weekdays sit consistently above weekends.
- Station-level distribution is bimodal: a large cluster of fully-saturated stations and a long tail of under-used ones.
- ACN sessions deliver ~9 kWh over ~5.9 hrs on average — long dwell, modest energy (workplace pattern).
- Spatial / geo view is omitted as a known data limitation (station coordinate merge unresolved); temporal and station-level findings stand.

AGENT 1 · DEMAND PREDICTION

# Forecasting utilization 1 hour ahead




## Model leaderboard

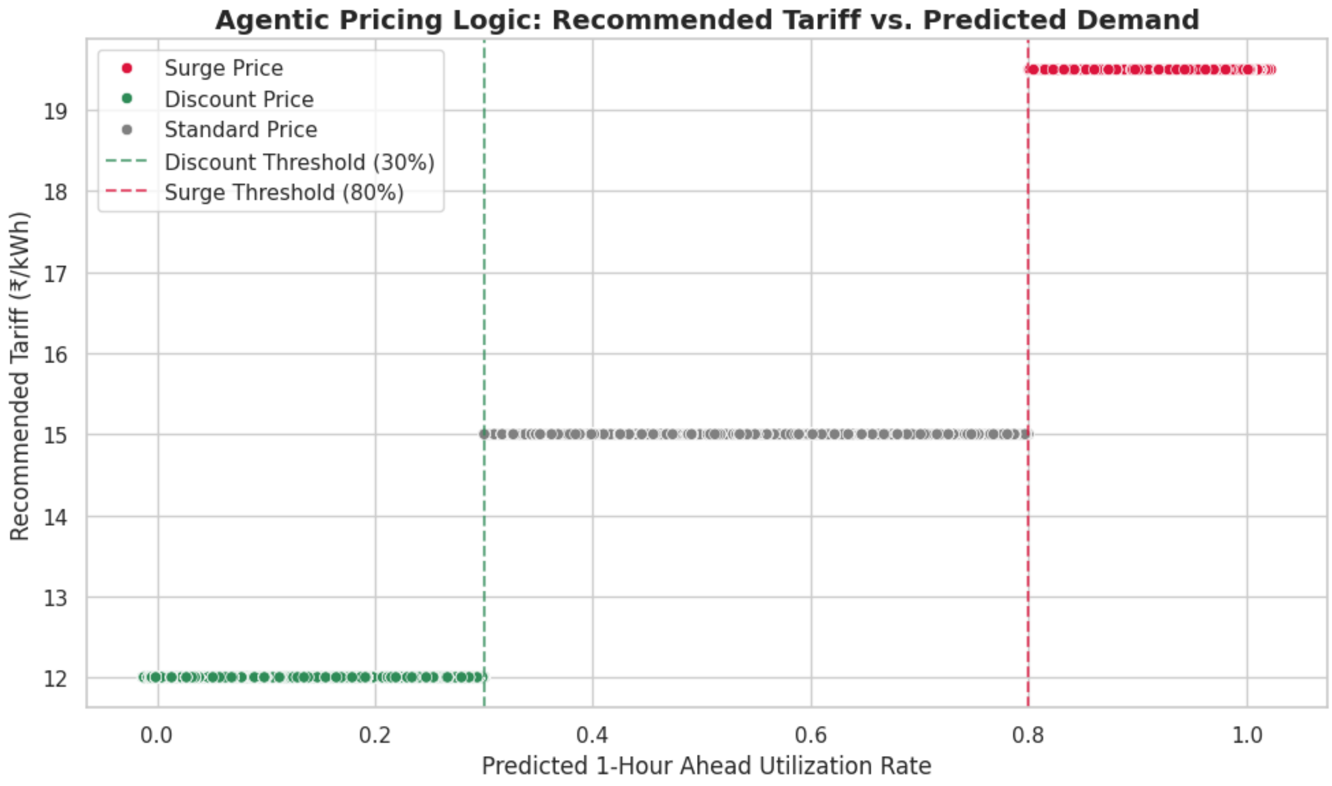
| Model | RMSE | $R^2$ |
|---|---|---|
| Grad. Boosting | 0.079 | **0.942** |
| Random Forest | 0.080 | 0.942 |
| Linear Reg. | 0.296 | 0.201 |

**Setup.** Time-ordered 80/20 split (no leakage). 10 temporal + operational features; target shifted −12 intervals (1 hr).

**Outputs.** Predicted utilization, congestion probability (≥80%), expected load.

*Tree ensembles capture the temporal structure; residuals are tight and centred around zero.*

# From forecast to dynamic ₹/kWh tariff




## Pricing rules engine

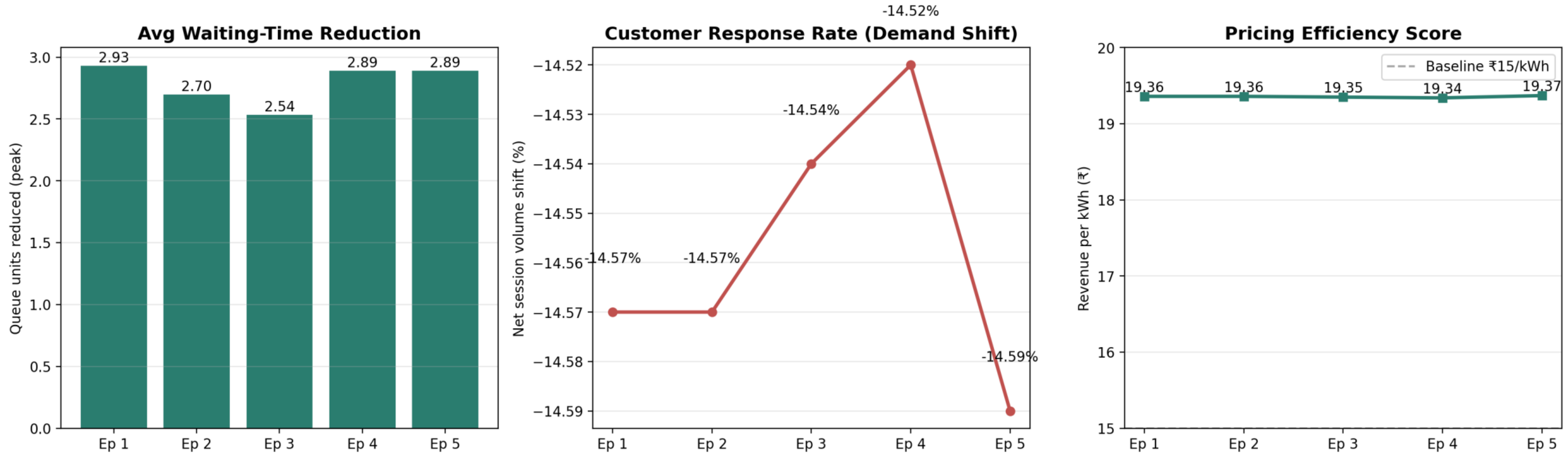
| | |
|---|---|
| **Surge** | Utilization ≥ 80%<br>**₹15 × 1.30 = ₹19.50** |
| **Standard** | 30% – 80%<br>**₹15.00 (baseline)** |
| **Discount** | Utilization ≤ 30%<br>**₹15 × 0.80 = ₹12.00** |

## Agent action mix (test set)

**Surge** 67.6% **Standard** 17.5%
**Discount** 14.9%

*Surge fires often because predicted utilization runs high (see limitations) — a recalibrated capacity baseline would rebalance this mix.*

# Feedback loop: revenue, queues & efficiency




## ≈ 2.8

Avg peak queue-proxy units reduced per episode

## −14.5%

Net session-volume shift (demand-elasticity proxy)

## ₹19.3+

Pricing efficiency (₹/kWh) — stable above ₹15 baseline

*Metrics are tracked over 5 sequential episodes; stability across episodes indicates a consistent, non-degrading policy under the current rule set.*

# Business, operational & policy takeaways

## Business

In the simulation, dynamic pricing is associated with ~10% higher revenue vs flat ₹15/kWh without new hardware — a software-only margin lever.

## Operational

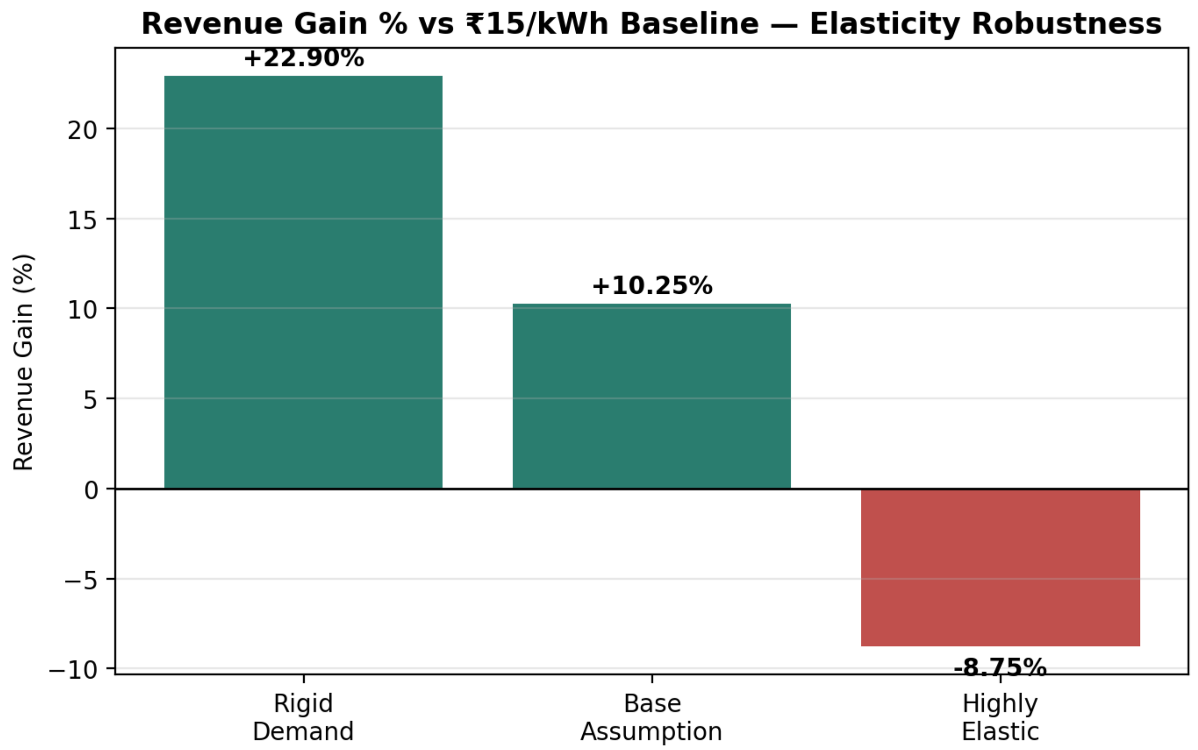
Surge/discount signals are associated with a smoother load curve — lower peak queues and more use of otherwise-idle off-peak capacity in simulation.

## Policy

Transparent, rule-based tariffs (capped surge, guaranteed off-peak discount) keep pricing explainable and consumer-fair.

**Recommended next step:** recalibrate the capacity baseline so the surge/standard/discount mix is balanced, then validate elasticity with a small live A/B pilot before full rollout.

# Revenue sensitivity to demand elasticity




## Reading the scenarios

**Rigid demand (users ignore price):** +22.9% — surge captures the most upside.

**Base assumption:** +10.25% — the headline, mid-elasticity case.

**Highly elastic (users react strongly):** −8.75% — surge backfires; discounts dominate. Pricing aggressiveness must be tuned to measured elasticity.

# Transparency: assumptions & limitations

## Key assumptions

- Charger draw 7.4 kW; baseline tariff ₹15/kWh.
- Surge +30% above 80% util; discount −20% below 30%.
- Elasticity assumed (surge −0.15, discount +0.25), not estimated from data.
- Forecast horizon fixed at 1 hour (12 × 5-min intervals).
- Queue-length proxy derived when raw traffic volume unavailable.

## Limitations

- Utilization runs high (~77% mean) and off-peak ≈ peak — capacity/occupancy ratio likely needs recalibration; this inflates surge frequency.
- Station geo-merge unresolved → spatial map empty.
- Revenue & elasticity are simulated, not from live A/B tests — directional, not causal.
- ACN and UrbanEV not fused at session level; insights reported per-source.

*No causal claims are made beyond what the simulation supports.*